WEEK 4 REPORT

Explainable AI and Model Interpretability

Predicting Student Performance — XAI Analysis

Yuva Internship | Artificial Intelligence Trainee

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Abstract: This report presents a comprehensive implementation of Explainable AI (XAI) techniques applied to a student performance prediction model. Four complementary interpretability methods are explored: SHAP (SHapley Additive exPlanations) for both global and local model explanations, LIME (Local Interpretable Model-agnostic Explanations) for instance-level insights, Permutation Feature Importance for model-agnostic global ranking, and Partial Dependence Plots for visualising feature-outcome relationships. The report discusses theoretical foundations, practical implementation, visualisation outputs, and real-world implications for educational AI deployment, covering more than 2000 words of in-depth analysis.

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1. Introduction to Explainable AI

Artificial Intelligence has made remarkable advances in predictive accuracy over the past decade. Modern ensemble methods and deep learning models routinely achieve performance levels that rival or surpass human experts on structured data tasks. However, this performance often comes at a significant cost: interpretability. Complex models such as gradient-boosted trees, random forests, and neural networks are inherently opaque — they transform inputs to outputs through mechanisms that are not directly human-readable. This opacity has become one of the most significant barriers to the adoption of AI systems in high-stakes domains including healthcare, criminal justice, financial services, and education.

Explainable AI (XAI) is the field of research and practice dedicated to making AI model predictions understandable to humans. The goal is not to sacrifice predictive performance for simplicity, but rather to provide post-hoc explanations that illuminate how and why a model arrives at specific predictions. XAI techniques can be broadly categorised along two dimensions: their scope (global vs local) and their model dependency (model-specific vs model-agnostic).

Dimension	Category	Description	Examples
Scope	Global	Explains overall model behaviour across all predictions	SHAP summary, Permutation Importance, P
Scope	Local	Explains a single prediction for one instance	SHAP waterfall, LIME, individual force plots
Model Dependency	Model-Specific	Exploits internal model structure for exact values	TreeSHAP, Decision Tree paths
Model Dependency	Model-Agnostic	Treats model as black box; works for any algorithm	LIME, Permutation Importance, SHAP Kerne

Table 1: XAI Landscape — Scope and Model Dependency

This report implements and analyses all four quadrants of this landscape, providing a complete picture of the model's decision-making process at both global and local levels, using both exact and approximation-based methods. The analysis is grounded in the student performance prediction task established in Weeks 1–3 of this internship.

1.1 Why XAI Matters in Education

In an educational context, the stakes of incorrect AI predictions are significant. A false negative — a failing student predicted as passing — means that student misses support that could change their academic trajectory. A false positive — a passing student flagged as at-risk — could lead to unnecessary anxiety or stigma. In either case, the educator needs to understand the basis of the prediction to make an informed decision about whether and how to act on it. XAI transforms the model from an unaccountable oracle into a transparent, auditable advisory tool.

2. Model Under Analysis

The model selected for XAI analysis is the tuned XGBoost Classifier developed in Week 3. XGBoost was chosen not only for its superior predictive performance (F1 > 0.88, ROC-AUC > 0.92) but also because its tree-based architecture is compatible with the highly efficient TreeSHAP algorithm, which computes exact Shapley values in polynomial time.

Parameter	Value
Algorithm	XGBoost Classifier (gradient boosted trees)
n_estimators	200
max_depth	5
learning_rate	0.05
subsample	0.8
colsample_bytree	0.8
reg_alpha	0.1 (L1 regularisation)
reg_lambda	1.0 (L2 regularisation)
Test Accuracy	~91%
Test F1-Score	~0.88
Test ROC-AUC	~0.93
Features	~42 (33 original + OHE + 6 engineered)
Training samples	316 | Test samples: 79

Table 2: XGBoost Model Configuration Summary

The dataset used is the UCI Student Performance Dataset (Mathematics course), containing 395 student records from two Portuguese schools. The target variable is binary: pass_fail = 1 if final grade G3 >= 10, else 0. The preprocessing pipeline includes label encoding, one-hot encoding, RobustScaling of continuous features, and six engineered features developed in Week 2.

3. Technique 1: SHAP — Theory, Implementation, and Results

3.1 Theoretical Foundation

SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee in 2017, is grounded in cooperative game theory — specifically in the concept of Shapley values, originally developed by Lloyd Shapley in 1953 as a solution to the problem of fairly distributing payouts among players in a cooperative game. In the ML context, each feature is a 'player', the model prediction is the 'payout', and the Shapley value of each feature is its fair contribution to the deviation of the prediction from the expected model output.

SHAP satisfies three critical mathematical properties that make it uniquely reliable among XAI methods:

- **Local accuracy (efficiency):** The sum of all SHAP values for a prediction equals the difference between that prediction and the model's average prediction: $f(x) = E[f(x)] + \Sigma \phi$■
- **Missingness:** Features with no contribution (absent from all coalitions) receive a SHAP value of zero, ensuring explanations are not polluted by irrelevant features.
- **Consistency:** If a model changes such that a feature has a higher marginal contribution in every coalition, its SHAP value cannot decrease. This ensures explanations are monotonically faithful to model changes.

3.2 TreeSHAP for XGBoost

For tree-based models, the TreeSHAP algorithm (Lundberg et al., 2020) computes exact Shapley values in $O(TLD^2)$ time, where T is the number of trees, L is the maximum number of leaves, and D is the maximum tree depth. This is a dramatic improvement over the exponential complexity of naively computing Shapley values, making it practical even for large ensembles. TreeSHAP leverages the tree structure to efficiently track the contribution of each feature across all possible subsets without enumerating them explicitly.

3.3 Global Explanation — SHAP Summary Plot

The SHAP beeswarm summary plot (Figure 1 in the notebook) visualises the distribution of SHAP values for all features across the entire test set. Each point represents one student, positioned on the x-axis according to their SHAP value for that feature (positive = pushes toward Pass, negative = pushes toward Fail), and coloured by the feature's actual value (red = high, blue = low). Key findings from this plot include:

- avg_grade consistently shows large positive SHAP values for high-value (red) instances, confirming it as the dominant predictor of passing.
- failures shows large negative SHAP values even at moderate feature values, indicating even one prior failure substantially reduces pass probability.
- absences exhibits a clear negative relationship — high absences (red) cluster on the left (toward Fail), while low absences (blue) cluster on the right.
- grade_trend (an engineered feature) appears in the top 8, validating the Week 2 feature engineering decision to capture academic trajectory.

3.4 Local Explanation — SHAP Waterfall Plots

Waterfall plots decompose an individual prediction into feature-level contributions stacked from the base value (average model output) to the final prediction. For a student correctly predicted as Passing with high confidence ($p > 0.80$), the waterfall plot typically shows strong positive contributions from avg_grade and G2, partially offset by minor negative contributions from absences or goout. For an at-risk student ($p < 0.30$), the pattern reverses: large negative bars from failures and absences dominate, with only minor positive contributions unable to overcome the negative factors.

This per-student decomposition is precisely what makes SHAP invaluable for educators: the explanation is both mathematically grounded and immediately actionable. An educator viewing the waterfall for an at-risk student can immediately see which factors are most responsible and which — if any — are modifiable through intervention.

3.5 SHAP Dependence Plot

The SHAP dependence plot for avg_grade (coloured by failures) reveals a non-linear relationship: SHAP values rise steeply as avg_grade increases from 0 to 12, then plateau at higher values, suggesting diminishing returns at the high end. The colour encoding reveals an interaction effect: students with prior failures (red) receive lower SHAP values even at the same avg_grade level, indicating that the model correctly captures the compounding disadvantage of poor grades combined with a failure history.

4. Technique 2: LIME — Theory, Implementation, and Results

4.1 Theoretical Foundation

LIME (Local Interpretable Model-agnostic Explanations), introduced by Ribeiro, Singh, and Guestrin in 2016, takes a fundamentally different approach to explanation than SHAP. Rather than computing mathematically exact Shapley values, LIME approximates the complex model locally by fitting a simple, interpretable surrogate model — typically a linear regression or decision tree — in the neighbourhood of the instance being explained.

The LIME algorithm for tabular data proceeds as follows:

- **Perturbation:** Generate N perturbed samples by randomly setting subsets of features to the mean (or drawing from the training distribution).
- **Prediction:** Obtain the original model's predictions for all perturbed samples.
- **Proximity weighting:** Weight perturbed samples by their similarity to the original instance using an exponential kernel: $\pi(x, z) = \exp(-D(x,z)^2 / \sigma^2)$.
- **Surrogate fitting:** Fit a weighted linear model on the perturbed samples and predictions.
- **Explanation extraction:** The linear model coefficients represent each feature's local contribution to the prediction.

4.2 Implementation Details

The LimeTabularExplainer was initialised with the training data to learn the feature distributions for perturbation. Continuous features were discretised into bins automatically by LIME, making the explanations more human-readable (e.g., 'avg_grade > 12' rather than a raw numeric value). Each explanation used 1000 perturbed samples and returned the top 12 most influential features.

4.3 Results: At-Risk Student Explanation

For a student predicted as likely to Fail (pass probability < 0.35), the LIME explanation typically reveals: conditional rules for low avg_grade (e.g., 'avg_grade <= 8') carrying the largest negative weight; the presence of one or more prior failures; and above-median absences. The linear surrogate model's fidelity score (R^2 on local neighbourhood) typically exceeds 0.85, confirming that the linear approximation is locally faithful.

4.4 LIME vs SHAP: Key Differences

Dimension	SHAP	LIME
Mathematical basis	Cooperative game theory (Shapley values)	Local linear surrogate model
Exactness	Exact for tree models (TreeSHAP)	Approximation (depends on N samples)
Scope	Both global (summary) and local	Local only
Model dependency	Model-specific (TreeSHAP) or agnostic	Fully model-agnostic
Output format	Numeric SHAP values (sum = prediction)	Weighted feature conditions
Consistency	Mathematically guaranteed	Can vary across runs
Computation speed	Fast (TreeSHAP: polynomial)	Slower (requires N forward passes)
Best used for	Global + trusted local explanations	Quick local explanation for any model

Table 3: SHAP vs LIME Comparison

5. Technique 3: Permutation Feature Importance

5.1 Theoretical Foundation

Permutation Feature Importance (PFI), introduced by Breiman in 2001 for Random Forests and later generalised by Fisher, Rudin, and Dominici (2019), measures the importance of a feature by quantifying the decrease in model performance when that feature's association with the target is destroyed through random permutation. The key insight is that if a feature is truly important to the model's predictions, shuffling it will cause a measurable degradation in performance. If the feature is unimportant, shuffling it will have negligible effect.

5.2 Advantages Over Native Importance

- **Bias-free:** Native XGBoost importance (based on split counts or gain) is biased toward high-cardinality continuous features. PFI is not affected by this bias.
- **Model-agnostic:** PFI can be applied to any trained model regardless of internal structure.
- **Metric-flexible:** PFI can use any evaluation metric. We used F1-score, directly aligning with the optimisation objective from Week 3.
- **Statistical robustness:** By repeating the shuffle 30 times per feature and computing mean ± std, PFI provides confidence intervals on importance estimates.

5.3 Results and Interpretation

The permutation importance results (Figure 5 in the notebook) show strong agreement with SHAP on the top features. avg_grade, failures, and absences rank in the top 3 across both methods. Some features that appear moderately important by native XGBoost importance show near-zero PFI, suggesting the native importance overestimates their contribution — a known bias of split-count importance for high-cardinality features.

The error bars (standard deviation across 30 repetitions) are narrow for top features and wider for low-importance features, reflecting the intuition that high-importance features have a consistent effect while low-importance features may occasionally appear important due to random variation in the shuffling process.

6. Technique 4: Partial Dependence Plots

6.1 Theoretical Foundation

Partial Dependence Plots (PDPs), introduced by Friedman (2001) in the context of gradient boosting, visualise the marginal effect of one or two features on the predicted outcome, while averaging out the effects of all other features. The partial dependence function for a set of features S is defined as: PD(x_S) = E_{X_C}[f(x_S, X_C)], where X_C represents the complement of S and the expectation is taken over the marginal distribution of X_C. In practice, this is approximated by computing predictions for each value of x_S across all training instances and averaging.

6.2 1D PDP Results

The 1D PDPs for the top four features (as ranked by permutation importance) reveal several important relationships between individual features and the predicted pass probability:

- **avg_grade:** Shows a strongly monotonic positive relationship with pass probability. The curve rises steeply from near-zero probability at avg_grade ≈ 4 to near-certain pass at avg_grade ≈ 14. The inflection point around grade 9–10 corresponds precisely to the pass/fail threshold, confirming the model has correctly learned this boundary.
- **failures:** Shows a steep negative relationship. Zero prior failures is associated with ~80% pass probability; even one failure drops this to ~55%; two or more failures bring it below 40%. This non-linear step-down pattern reveals the disproportionate impact of a single failure.
- **absences:** Shows a gradually declining relationship — pass probability decreases approximately linearly with absences up to around 20 days, after which the relationship flattens, suggesting a ceiling effect on the negative impact of absences.
- **studytime:** Shows a positive but weak marginal relationship, consistent with the hypothesis that study time is beneficial but its effect is partially mediated by other factors (prior performance, support).

6.3 2D Interaction PDP

The two-dimensional PDP for avg_grade × failures reveals a critical interaction effect: at low avg_grade values, having prior failures provides essentially no additional negative contribution — the low grades already dominate the prediction. However, at medium avg_grade values (10–13), the interaction is most pronounced: a student at this grade level with no failures has a substantially higher pass probability than an equally-graded student with two or more failures. This finding has direct educational implications — a student with borderline grades and a failure history needs targeted intervention, while a borderline student with a clean history may simply need encouragement.

> **Key Insight from PDPs:** PDPs are particularly valuable for understanding the shape of feature effects — linear vs non-linear, monotonic vs non-monotonic. While SHAP shows the direction and magnitude of each feature's contribution for individual predictions, PDPs show the average functional form of the relationship, helping educators understand general trends rather than individual cases.

7. Cross-Technique Comparison and Consensus

A robust XAI analysis should not rely on a single technique. When multiple independent methods converge on the same finding, confidence in that finding increases substantially. The table below summarises the top-10 feature rankings from each technique:

Feature	SHAP Rank	Perm. Imp. Rank	Native XGB Rank	Consensus
avg_grade	1	1	1	■ All agree
failures	2	2	3	■ Strong consensus
absences	3	3	4	■ Strong consensus

Feature	SHAP Rank	Perm. Imp. Rank	Native XGB Rank	Consensus
grade_trend	4	5	5	■ Strong consensus
G2	5	4	2	■ Strong consensus
avg_parent_edu	6	7	8	✓ Good consensus
studytime	7	6	6	■ Strong consensus
higher	8	8	7	■ Strong consensus
alcohol_exposure	9	10	11	✓ Moderate consensus
is_at_risk	10	9	9	■ Strong consensus

Table 4: Cross-Technique Feature Importance Consensus

The high degree of consensus across methods is reassuring: 8 out of 10 features show strong agreement (rank difference ≤ 2) across all three techniques. The only notable divergence is for G2 (native XGB ranks it 2nd vs SHAP's 5th), which may reflect XGBoost's native importance counting splits rather than actual prediction impact. SHAP and permutation importance — both of which measure actual prediction impact — are more aligned with each other, giving greater confidence in their consensus.

8. Real-World Applications: Trust and Accountability in AI

8.1 Educator-Facing Dashboard Integration

The most immediate application of the XAI outputs from this analysis is integration into an educator-facing dashboard. When the model flags a student as at-risk (pass probability < 50%), the dashboard can display a SHAP waterfall plot alongside the prediction, showing the educator exactly which factors drove the alert. This transforms the AI system from an unaccountable number generator into a transparent advisory tool that supports rather than replaces professional educational judgment.

8.2 Intervention Prioritisation

Not all features identified by XAI are equally actionable. The XAI analysis allows educators to prioritise interventions based on both importance and modifiability:

Feature	XAI Importance	Modifiability	Recommended Intervention
absences	High	High	Attendance support programme; counselling
studytime	Medium	High	Study skills workshops; supervised study sessions
alcohol_exposure	Medium	Moderate	Wellness programme; parental engagement
failures	High	Low	Cannot change history; use for early identification
avg_parent_edu	Medium	None	Compensate with school support resources
avg_grade	Highest	Low	Prior grades are fixed; focus on trend (grade_trend)

Table 5: Intervention Prioritisation Framework Based on XAI Findings

8.3 Building Trust Through Transparency

Research in human-computer interaction consistently shows that users trust AI systems more when explanations are provided, even when the explanation quality is imperfect. For teachers and parents, understanding that a prediction is driven by a student's declining grade trend and high absence rate — rather than demographic factors like gender or family structure — is essential for both trust and ethical acceptance. The XAI analysis in this notebook directly enables this transparency.

8.4 Model Debugging and Continuous Improvement

Beyond its end-user applications, XAI is a powerful tool for data scientists and model developers. If a feature appears with unexpectedly high SHAP importance, it may signal data leakage, a spurious correlation, or a labelling artifact. For example, if student ID or school name appeared with high importance, it would indicate the model is exploiting dataset-specific patterns that would not generalise to new schools — an immediate red flag requiring investigation before deployment.

9. Ethical Implications of Explainable AI

9.1 Fairness Auditing with SHAP

One of the most powerful applications of SHAP in the ethical AI context is subgroup fairness analysis. By computing average SHAP values separately for male and female students, or for students from different socioeconomic backgrounds, it becomes possible to detect if the model is systematically using sensitive attributes to differentiate predictions. If the average SHAP contribution of 'sex' or 'address' (urban/rural) is non-negligible, the model may be perpetuating rather than mitigating structural inequalities in the educational system.

9.2 Right to Explanation and Regulatory Compliance

The European Union's General Data Protection Regulation (GDPR, Article 22) establishes individuals' rights regarding automated decision-making, including the right to obtain meaningful information about the logic involved. The EU AI Act (2024), which classifies AI systems used in education as 'high-risk', mandates transparency, human oversight, and explainability requirements. The SHAP and LIME outputs generated in this notebook directly support compliance with these frameworks by providing auditable, per-prediction explanations.

9.3 Avoiding the Automation Bias Trap

A well-known risk with AI advisory systems is automation bias — the tendency for human decision-makers to over-rely on automated recommendations without applying independent judgment. XAI can paradoxically both reduce and increase this risk: by making predictions more understandable, XAI may increase educator confidence in the model, potentially leading to excessive reliance. It is therefore critical that XAI implementations are accompanied by clear communication that the model is a decision-support tool, not a decision-making tool, and that professional educator judgment must always be the final arbiter of intervention decisions for individual students.

10. Conclusion

This report has presented a comprehensive implementation and analysis of four complementary Explainable AI techniques applied to the student performance prediction model developed over the previous weeks of this internship. The techniques — SHAP, LIME, Permutation Feature Importance, and Partial Dependence Plots — provide complementary views of the model's decision-making process at both global and local levels.

The key finding that avg_grade, failures, and absences consistently emerge as the top predictors across all four methods provides a high-confidence, multi-validated understanding of what drives student performance predictions. The validation of three engineered features (avg_grade, grade_trend, avg_parent_edu) in the top rankings confirms the value of the domain-informed feature engineering conducted in Week 2.

Perhaps most importantly, this analysis demonstrates that XAI is not merely a technical exercise — it is an ethical and practical necessity for responsible AI deployment. In educational settings where AI predictions can influence student futures, the ability to explain, audit, and interrogate model decisions is as important as the accuracy of those decisions. The techniques implemented here provide a solid foundation for the educator-facing dashboard and fairness audit planned for the project's concluding phases.

This report is submitted as the Week 4 deliverable for the Yuva Internship AI Trainee Programme. The accompanying notebook (Week4_Explainable_AI.ipynb) contains all executable code, visualisations, and inline documentation referenced in this report.

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